Asansol Engineering College

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Name : Rahul Kumar

Roll no : 10800123147

Dept. : CSE

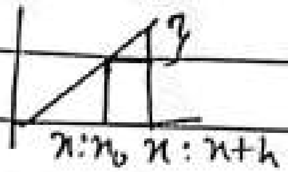
Year : 3rd

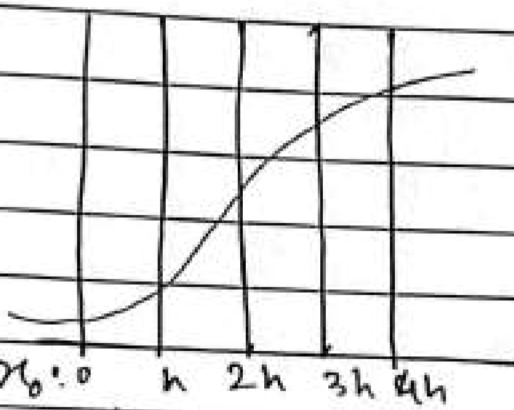
Subject : Numerical method

Sub. Code : OEE - IT 601 A

Numerical Analysic

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$$





Finite differences

Grid → h

Approximate value (Reasonable solution)

Δ forward diff operation

$\Delta f(x) = f(x+h) - f(x)$ Current $= \Delta y_i = y_{i+1} - y_i$

Next

$\Delta\Delta f(x) = \Delta[f(x+h) - f(x)]$

A linear operation $T(f+g) = T(f) + T(g)$.

$\therefore \Delta^2 f(x) = \Delta f(x+h) - \Delta f(x)$

$= f(x+2h) - f(x+h) - f(x+h) + f(x)$

$= f(x+2h) - 2f(x+h) + f(x)$

$\nabla f(x) = f(x) - f(x-h)$

Current − prev

$\nabla^2 f(x)$ -

Shift operation:

$$E\,f(x) = f(x+h)$$
$$E^2 f(x) = f(x+2h)$$
$$E^3 f(x) = f(x+3h)$$
$$E^4 f(x) = f(x+4h)$$
$$E^n f(x) = f(x+nh)$$
$$E^{-1} \cdot f(x) = f(x-h)$$
$$E^{1/2} f(x) = f(x+\tfrac{1}{2}h)$$

Only rational power

$$\Delta f(x) = f(x+h) - f(x)$$
$$= E f(x) - f(x)$$
$$= (E-1) f(x)$$
$$\Rightarrow \Delta = E - 1$$

$$\nabla f(x) = f(x) - f(x+h)$$
$$f(x) - E^{-1} f(x)$$
$$= (1 - E^{-1}) f(x)$$
$$\nabla = 1 - E^{-1}$$

$$\Delta\nabla f(x) = \Delta[f(x) - f(x+h)]$$
$$= f(x+h) - f(x) \cdot f(x) + f(x-h)$$
$$= E f(x) - 2 f(x) + E^{-1} f(x)$$
$$= (E + E^{-1} - 2) f(x)$$
$$\therefore \Delta\nabla = E + E^{-1} - 2$$

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Q. Find $\Delta^2(x^2-3x+5)$ at $x=3$ given $h=1$

$(x+1)^2 - 3(x+1) + 5 - x^2 + 3x - 5$

$x^2 + 2x + 1 - 3x - 3 + 5 - x^2 + 3x - 5$

$2x - 2 \mid x = 3$

$x = 4$

$\Delta(2x-2) = 2(x+1) - 2 - 2x + 2$

$= 2$

If $f(x)$ be a poly of degree n

$\Delta^n f(x) = K$

$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \cdots$

$= f(x) + hDf(x) + h^2 D^2 f(x) + \cdots$

$= \left(1 + hD + \frac{h^2 D^2}{2!} + \frac{h^3 D^3}{3!} + \cdots\right) f(x)$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots$

$Ef(x) = e^{hD} f(x)$

$E = e^{hD}$

$1 + \Delta = e^{hD}$

taking log

$\frac{1}{h} \log(1+\Delta) = D$

$Df(x) = \frac{1}{h} \log(1+\Delta) f(x)$

$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} \cdots$

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$y_h \{\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - - -\} f(n)$

$\Delta = E - 1$

$E = 1 + \Delta$

$E^n f(x) = f(x + nh) = (1 + \Delta)^n f$

$E^n = (1 + \Delta)^n = \sum_{r=0}^{n} {}^nC_r \Delta^r = E^n$

$(1+x)^n = {}^nC_0 + {}^nC_1 x - {}^nC_2 x^2 - - \cdot {}^nC_n x^n$

$= \sum_{r=0}^{n} \cdot {}^nC_r x^r$

Ex: if $f(0) = 1$

$f(1) = 3$

$f(2) = 8$

$f(3) = 13$

find $f(6) = f(0 + 6 \cdot 1)$

$(1+\Delta)^6 f(0)$

$= 18$

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